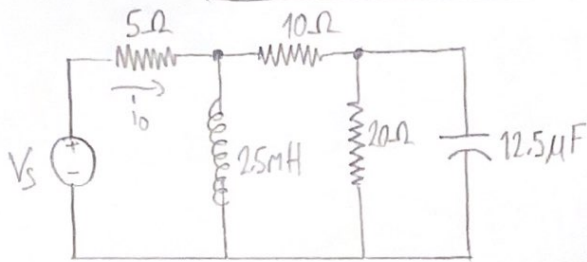


EE 201 - Assignment # 3

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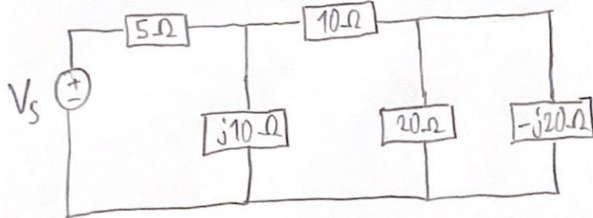


$$V_s = 20 \sin(4000t) \text{ V}, \quad \omega = 4000 \text{ rad/sec}$$

$$Z_{5R} = 5 \Omega, \quad Z_{10R} = 10 \Omega, \quad Z_{20R} = 20 \Omega$$

$$Z_L = j4000 \times 25 \times 10^{-3} = j10 \Omega, \quad Z_C = \frac{1}{j4000 \times 12.5 \times 10^{-6}} = -j20 \Omega$$

Transform passive elements to impedances



$$V_s = 20 \sin(4000t) = 20 \cos(4000t - 90^\circ)$$

$$\underline{V}_s = 20 \angle -90^\circ$$

$$I = \frac{\underline{V}_s}{Z_{eq}} = \frac{20 \angle -90^\circ}{10\sqrt{2} \angle 45^\circ} = \sqrt{2} \angle -135^\circ$$

$$Z_{eq} = ((20 \parallel -j20) + 10) \parallel j10 + 5$$

$$= \left(\frac{(-j20) \cdot 20}{20 - j20} + 10 \right) \parallel j10 + 5$$

$$= \left(\frac{-j20 - j10 + 10}{1 - j} \right) \parallel j10 + 5$$

$$= \frac{30j - 10}{2 + 2j} + 5 = \frac{40j(2 - 2j)}{8} = 10j + 10$$

$$Z_{eq} = 10j + 10 = 10\sqrt{2} \angle 45^\circ$$

$$I = \sqrt{2} \angle -135^\circ \Rightarrow i_o(t) = \sqrt{2} \cos(4000t - 135^\circ) \text{ A}$$